

Anomaly Math

Logan Joseph

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1 Quick Mathematical Dictionary

We want to use the language of forms for convenience. Each object we work with normally in QFT is really the coefficients of a form expressed in some coordinate basis. But things get much simpler when working with the geometric object and forgoing a specific local representation. The basic mathematical building blocks for gauge theory are lie algebra valued forms:

- Connection 1 Form: $A = A_\mu dx^\mu = A_\mu^a T^a dx^\mu$
- Curvature 2 Form: $F = dA + A \wedge A = \frac{1}{2} F_{\mu\nu} dx^\mu \wedge dx^\nu = \frac{1}{2} (\partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu]) dx^\mu \wedge dx^\nu$

As the notation implies, A is the gauge field and F is the field strength. Physically, the gauge field A_μ accounts for the junk you get when taking the derivative of a field when applying a gauge transformation. I.e. its something like $\partial_\mu \alpha$ since $\partial_\mu (e^{i\alpha(x)} \phi(x))$ picks up an extra phase term. Mathematically, it is describing how to have a consistent notion of transporting local objects like vectors along curves when your underlying space is curved. When its nice and flat the normal derivative does just fine, but when it isn't, you have a non trivial A that allows you to define a covariant notion of taking a derivative, given by

$$D = d + A$$

Essentially using this D cancels out the annoying phase term. Physically, F is the field strength, and geometrically it measures the failure of geometric quantities to be preserved when you transport things around a closed loop. This ultimately encodes the measurable physical fields, and to see why we consider how these objects behave under a local gauge transformation. (The following is understood formally using the notion of principal G bundles as that allows you to formulate how to attach a gauge group G to your spacetime M , but I don't think its insightful to delve into that for the purpose of this talk) Under a local gauge transformation $U \in G$, we have

$$A = UAU^{-1} - (dU)U^{-1}$$

$$F = U F U^{-1}$$

We can now see why F encodes the physical data. Constructing Lagrangians out of polynomials of A might pick up extra terms which spoil gauge invariance (which of course we demand). However, constructing our Lagrangians out of traces of F makes things manifestly gauge invariant cause you just use the cyclicity of the trace to move the U 's and the cancel them.

2 Gauge Theory and Matter in a Mathematical Language

Lets now take a big step back and start from the beginning, how do we formulate this all in a purely geometric way and where does matter come into play? We start with our spacetime M . In the context of physics, we care about smooth manifolds as our spacetime, in QFT, this is just our Minkowski space. We also consider the gauge group underlying the symmetry we want to consider. This is a Lie Group G . Now how do we get these to interact? In QFT we watch as G acts on our fields (which are constructed from representations of G , and are functions of our spacetime M) and gives us a new phase shifted field. Mathematically we need bundles to describe this in a global, geometric way. This can get a bit complicated, but we can avoid the most general theory by restricting the use case to gauge theory which is what we are interested in. The construction is as follows

1. Start with three different manifolds (yes three), call them P , M , and G . In our use case M is spacetime, G is our symmetry group, and P is called a principal bundle if the following points are true
2. There exists a projection map $\pi : P \rightarrow M$ which satisfies a local triviality condition. This means that for a local neighborhood $U \subset M$, then there is a homeomorphism $\phi : \pi^{-1}(U) \rightarrow U \times G$.
3. There exists a continuous right action $P \times G \rightarrow P$ that preserves the fibers of P . I.e. if $y \in P_x$, then $gy \in P_x$ for all $g \in G$. This is just a formal way of describing the gauge transformation itself

Lets break this down a little bit. The local triviality condition on the projection map essentially means that we can think of a principal bundle P as **LOCALLY** looking like $M \times G$. If P is literally equal to $M \times G$ then this bundle is considered trivial. In other words, it can locally look like this product (i.e. a copy of G attached to every point in spacetime) but globally there could be some weird twisty stuff going on that makes it not globally just a product of our spacetime M and symmetry group G . So why is this an object we want to work with? In QFT, we think of a field $\psi(x)$ and then its gauge transformed version $\psi'(x)$ as yielding the same physics at the end of the day. In other words, at a point x , there are a bunch of different fields ψ that all give the same physics, and are related to one another via a gauge transformation. The fibers of our principal bundle at a point $x \in M$ essentially encode all of those ψ 's. And the third point

above is just the mathematical language of how we move around in a fiber to different ψ 's by doing a gauge transformation. A and F are objects naturally baked into this construction. A is a connection 1 form on this bundle (A geometrically allows us to transport between different fibers), and F is the curvature 2 form on this bundle.

As a quick aside before we continue, at this point in the mathematical construction we have pure gauge stuff and all we have is our field A . Note we have not incorporated matter yet and this might seem a bit flipped from physics, where normally we right down the matter and then "discover" what our gauge symmetry is and then construct a specific A/F . However this construction sees the gauge stuff as the starting point and then the next step (if desired) is to add matter content.

OK, now time to add matter. Remember that in QFT, matter lives in some random representation of our gauge group that happens to match experiment. For example, quarks live in the fundamental of $SU(3)$, meaning that there is some internal 3D color vector space that the field lives in, and the components are evaluated at a spacetime point x . So once again, there seems to be two spaces we are working with, our spacetime M , and some internal vector space where our field content lives. This kind of mimics what we had above, and unsurprisingly we have a very similar construction, ultimately called the "associated vector bundle to P ". Lets break this down. First, a vector bundle in general you can just think of as the same construction above (minus the third point about gauge transformations), but instead of attaching a group G to each point in space, we attach a vector space V to each point in space. An example of this is a cylinder, we take a circle S^1 and trivially attach $\mathbb{R}^1$ to each point. This is a trivial vector bundle. Or, a non trivial vector bundle constructed from the same pieces is the mobius strip, which looks like $S^1 \times \mathbb{R}^1$ **LOCALLY**, but globally there is a twist where you end up facing backwards when you loop around once.

This "associated vector bundle to P ", call it E , is a specific type of vector bundle and its construction is as follows:

1. Start with a principal bundle P , constructed from your spacetime M and gauge group G .
2. Choose a particular linear representation $\rho : G \rightarrow GL(V)$ of your group over some desired vector space (usually the complex numbers).
3. Now, construct $E = P \times_{\rho} V \equiv (P \times V)/G$ with fibre V . E is your associated vector bundle!

Lets elaborate on these points since this seems a bit random. Our principal bundle G doesn't have any matter to act our A on quite yet, so we need to form that. We choose a particular representation (for example the trivial rep of $SU(3)$ if we desired to describe natures quarks) of our G over the vector space (which from now on Ill just say is the complex numbers) and say that our matter lives in that rep. Intuitively, we can think of the third point as attaching the complex numbers to each point in space, and then inheriting the

data from P to allow us to apply our gauge transformations (elements of G) to our matter (representations of G over the complex numbers). Now, the connection on this bundle is just the same as from the principal bundle, but were we swap out the abstract A and use the specific generators of that rep, i.e. $\nabla = d + \rho(A)$. Note that in the physics language this point is sort of occluded when we write $D_\mu = \partial_\mu + A_\mu$ because it doesn't really indicate if we have chosen a particular rep yet, or if so, what rep we did choose. The fields themselves are "sections" Γ of this vector bundle. A section is a map which maps M to E by taking a slice of E . Essentially its writing down what $\psi(x)$ is. A flat slice through E would just be a 0 field, but if the slice has some curves that would mean the vector in field space is oscillating as you change x . Overall we can see that having E makes the notion of choosing a representation to include matter and do our QFT precise.

Great, now we can start. Remember that all of this machinery was to define the same stuff we do in QFT, but in a global and geometric way. I.e. if I tried doing perturbative QFT in the context of the cylinder vs mobius strip, Id get the same stuff since they locally are just $S^1 \times \mathbb{R}^1$. However, that misses the fact that the mobius strip has some non trivial topological winding to it. This construction allows us to not lose the topological info like we would otherwise in perturbative QFT.

Also notice that this construction also signals how anomalies are fundamentally something that arise from the pure gauge part of the theory, and how incorporating a specific type of matter is just one way to feel it. But at the end of the day, the anomaly is there whether or not we do QFT at the high energy quark level, or at the low energy pion EFT level. Personally I like this perspective cause the physics way of defining anomalies (at least to me) made it seem like they were a result of the matter you chose, not a fundamental geometric obstruction which lives in the gauge theory.

3 Why Does Perturbative QFT not See Topology?

As we will see here, it is by construction that the perturbative QFT techniques we are used to essentially make everything trivial and thus ignore any potential topological facts.

At the level of the Feynman path integral,

$$Z = \int \mathcal{D}\mathcal{A} e^{iS_M}$$

(where S_M is the action on Minkowski space) everything is exact and all the topological information is there, since the measure tells us we need to integrate over all possible A 's, both the configurations which have no interesting topology and the configurations which do. Lets also do a couple quick math tricks to help classify this nicely. First, lets Wick rotate, so now we have

$$Z = \int \mathcal{D}\mathcal{A} e^{-S_E}$$

where S_E is now the action over $\mathbb{R}^4$ Euclidean space. Now, we know that to have a finite action, we need our physical quantity F to vanish at the boundary. Mathematically, we can describe this by essentially pinching the $SU(3)$ boundary at infinity into a point, which turns $\mathbb{R}^4$ into S^4 . Now the punchline: we are now working on a compact manifold, S^4 , and we physically care about gauge groups $G = SU(N)$ and that allows us to classify everything via homotopy theory. In other words we just open up a book on algebraic topology and cite that, (Elliot mentioned this last week), $\pi_3(SU(N)) = \mathbb{Z}$! Therefore, all of the interesting topology that can exist is classified via an integer, k . In other words, each choice of configuration of A means we have implicitly decided to work in a principal bundle whose construction is classified by some integer k . Schematically, we can then write out the path integral as

$$Z = \sum_{k=-\infty}^{\infty} \int \mathcal{D}\mathcal{A}_k e^{-S_E[A_k]}$$

where we integrate over all the possible A 's for each possible principal bundle which in this case is classified by an integer k !

If you want, you can ignore everything I just said and remember the fact that for theories of interest, i.e. $SU(N)$ gauge theories in our normal spacetime, all the interesting topology can be classified by an integer, depending on what gauge configuration you chose. Specifically $k = 0$ means everything is trivial and there is no interesting winding or topology, and $k \neq 0$ means there is interesting topology! Lets see how normal QFT methods not only implicitly choose $k = 0$, but also confine us to work entirely in the $k = 0$ bundle meaning that we entirely miss everything that is not $k = 0$!

As stated before, the physical requirement is that $F = 0$ at the boundary, i.e. we cant measure an electric field out at infinity. However, if you follow the path of a standard QFT book, you tackle the path integral by claiming that $A = 0$ at the boundary (and so it can be smoothly deformed to 0 everywhere) and look at fluctuations around 0. What this does it make the lower bound of our action 0. This gives us our standard perturbative techniques and Feynman diagram methods where everything is ordered in terms of some perturbative parameter g . However, this is an over assumption! As we saw in Elliot's talk, we can choose pure gauge configurations, $A \sim h^{-1}dh$. These give instanton solutions, and it is famously known that for a single instanton ($k = 1$), the minimum energy configuration is something like

$$S_{instanton} \geq \frac{8\pi^2}{g^2}$$

This is because instantons are a non perturbative topological feature that are sourcing some energy in the system. Ok but lets actually see where this comes from by skipping the formalism for a sec and reminding ourselves of Elliot's talk(?), where we saw that $Tr(F \wedge F)$ encodes the topological data, and that the integer k is expressed as $k = \frac{1}{8\pi^2} \int Tr(F \wedge F)$. In differential forms the Yang-Mills action is given by $S = \frac{1}{g^2} \int Tr(F \wedge \star F)$. For two differential forms, A, B , of the same degree, their analog of a dot product

(scalar, positive definite) is $\int Tr(A \wedge \star B)$. This is because the integrand is essentially proportional to the volume form, and the trace traces down all the group matrices to give a number. Anyways, we exploit the fact that it is positive definite to write

$$\begin{aligned} \int Tr((F \mp \star F) \wedge \star(F \mp \star F)) &\geq 0 \\ \int Tr(F \wedge \star F \mp F \wedge \star \star F \mp \star F \wedge \star F + \star F \wedge \star \star F) &\geq 0. \end{aligned}$$

In 4D Euclidean space (where the metric is just $\mathbb{I}$), $\star\star = 1$, $\star F \wedge \star F = F \wedge F$, and $F \wedge \star F = \star F \wedge F$. Therefore

$$\begin{aligned} \int Tr(F \wedge F) &\geq \mp \int Tr(F \wedge \star F) \\ 8\pi^2 k &\geq \mp g^2 S \\ S &\geq \frac{8\pi^2 |k|}{g^2} \end{aligned}$$

So, depending on which bundle structure we are in (again, characterized by k), that sort of determines what we should be expanding about, and normal perturbative methods expand around 0, i.e. $k = 0$.

4 Constructing Topological Invariants via Characteristic Classes

Again, we have seen how to formulate the geometry here, and obviously the topology is encoded somewhere in the structure of these bundles, but that's not useful unless we can actually construct topological invariants via this construction to actually be able to study what is going on in our Lagrangians. Using homotopy theory showed us that we expect some integer classification of our 4D spacetime, and the integral I gave for that integer k above came out of nowhere. We are now going to see how characteristic classes allow us to measure essentially what is obstructing the ability to "untwist" some non trivial principal bundle into a trivial one, and allow us to rigorously construct formulas for these invariants.

To begin constructing topological invariants which affect the physics, we require two things:

1. Gauge Invariance
2. Invariance of the metric. This is so the terms are truly topological and a feature of the gauge geometry, and not some feature of how you decided to measure distances. This is what makes it "topological"

To quickly touch base with what we have done so far, the standard Yang Mills action has the metric in the integrand. This is what makes it dynamical and not just a feature of the geometry. As we have seen before, $Tr(F \wedge F)$ depends only on the Levi Civita symbol, but not the metric, this is what makes that term topological! So, we are on the hunt for writing down terms like the latter.

I thought I'd mention that characteristic classes and the role of cohomology within it is an incredibly rich and deep topic that we will only scratch the surface of for the use case of physics.

Our first step is constructing so-called "invariant polynomials" out of the Lie Algebra. The definition is simple, we build a function P which takes complex matrices and gives a complex number, and that satisfies

$$P(hXh^{-1}) = P(X)$$

for all $X \in \mathfrak{g}$ and $h \in G$ (I'm using h instead of g to avoid confusing with the Yang Mills coupling g).

We can extend this definition of P to be functions of $\mathfrak{g}$ valued p -forms. This is so that we can construct polynomials of F , which remember is a 2 form with coefficients in the Lie Algebra $\mathfrak{g}$. This is done by the following, if you have two matrix forms $X = X^1\eta_1$ and $Y = X^2\eta_2$, then we just define

$$P(X, Y) = \eta_1 \wedge \eta_2 P(X^1, X^2)$$

Where now on the right hand side P is just a function of matrices and you can use the standard definition. If this seems confusing, this just boils down to saying that if we have terms in a polynomial like $(X^a)^2$ (where X^a is a matrix) then we can just replace it with $X \wedge X$ (where X is a matrix form) and we have the same notion of an invariant polynomial. But again this is just so we can justify that it is formally meaningful to construct polynomials out of F (a Lie Algebra (matrix) 2 form) which is now exactly what we are going to do.

Remember, our curvature 2 form F transforms under a gauge transformation as UFU^{-1} whereas A transforms as $A = UA^{-1}U - (dU)U^{-1}$. That extra term in A ruins the construction of invariant polynomials but F is perfectly suited for it (physically, we expect this since F is the measurable object). Now, heres the punchline, traces work perfectly for this! Schematically

$$Tr(UFU^{-1}) = Tr(F)$$

since you can just permute the matrices. If you really wanted to be careful you'd have to use the above definition with forms and verify that this is actually how it ends up working but that's just pointless detail, hopefully you are convinced that the trace works here. You might also ask if there are any other polynomials, but some linear algebra theorems tell us that anything we can write down for P can be written as traces so actually this is all we have to care about!

Now, we begin to see where this might be going in terms of physical applications by citing the Chern-Weyl theorem which states that for an invariant polynomial $P(F)$:

1. $dP(F) = 0$
2. For two different curvature 2 forms F, F' , then $P(F') - P(F)$ is exact (it can be written as $d(\text{something})$).

Point 1 is powerful, it is telling us that these polynomials are closed and so when integrated over a manifold will yield a result that just corresponds to the topology. However, this polynomial is still sensitive to local fluctuations. If I take $A \rightarrow A + \delta A$, then F changes by a δF , and we could get a slightly different polynomial. The second point above is what allows us to define "characteristic classes". If the difference between two forms is exact, then we say they belong to the same De-Rahm cohomology class. Cohomology is a complex but rich mathematical idea that allows us to in a completely topological way characterize spaces like vector bundles. So, if the F from two vector bundles give P 's that differ only by an exact form, then they will have the same topological invariant! Because of point 2, we now have a way of constructing global topological invariants from our differential forms!

In general, characteristic classes are an entirely algebraic topology construction, so if you tried looking this up on Wikipedia it wouldn't look like what I'm doing at all. The Chern Weyl theorem helped us arrive at that idea via differential geometry and gave us an explicit construction for this particular use case. To summarize, we can organize all the Polynomials P into equivalence classes based on the De Rahm cohomology they live in. Choosing a different P in the same cohomology would end up giving the the same invariant after integration a la Chern Weyl.

Ok, so that's a really cool idea, but not very useful if we cant actually compute what these classes of polynomials actually are. Luckily, there is a way to do that! Lets start by defining this polynomial called the total Chern class

$$\begin{aligned}
c(F) &\equiv \det \left(I + \frac{iF}{2\pi} \right) \\
&= I + c_1(F) + c_2(F) + \dots \\
c_0(F) &= 1 \\
c_1(F) &= \frac{i}{2\pi} \text{Tr}(F) \\
c_2(F) &= \frac{1}{2} \left(\frac{i}{2\pi} \right)^2 (\text{Tr}(F) \wedge \text{Tr}(F) - \text{Tr}(F \wedge F)) \\
&\vdots \\
c_k(F) &= \left(\frac{i}{2\pi} \right)^k \det(F)
\end{aligned}$$

In the second line I expanded the determinant in a Taylor series, and got terms like $c_j(F)$ which are $2j$ forms. This means that if our spacetime is dimension n , then $c_a(F)$ if $2a > n$, so the series has to terminate at k , where k is the largest number such that $2k < n$. Note that this sum is a "formal sum", just a way of adding things of different degree, they are not "literally" being added together. These $c_k(F)$'s are called Chern classes, and because they are closed via the first point of the Chern Weyl theorem, they are elements of a De Rahm cohomology. In other words, this polynomial allows us to write out all the different

topological invariants based on F , and we don't need to worry about anything else because anything else will be equivalent (in the cohomology sense) to one of the $c_k(F)$'s after integrating. Therefore these Chern classes give us everything we need to know to construct topological invariants!

Lets see a very quick example of this. Consider a $\mathbb{C}^2$ bundle over our 4d spacetime M with $G = SU(2)$. (This would be saying to look at an associated vector bundle crafted from the fundamental representation of $SU(2)$, if I did adjoint then Id have $\mathbb{C}^3$, etc...). Then we can write $F = F^a(\sigma_a/(2i))$ where $F^a = \frac{1}{2}F_{\mu\nu}^a dx^\mu \wedge dx^\nu$. Then we get

$$\begin{aligned} c(F) &= 1 + \frac{1}{4} \left(\frac{i}{2\pi} \right)^2 (F^1 \wedge F^1 + F^2 \wedge F^2 + F^3 \wedge F^3) \\ c_0(F) &= 1 \\ c_1(F) &= 0 \\ c_2(F) &= -\frac{1}{16\pi^2} F^a \wedge F^a \\ c_{>2}(F) &= 0 \end{aligned}$$

But earlier I claimed that this represents some topological invariant that is physically meaningful, so lets see a couple of examples. First, you will have to take my word that integrals over these Chern classes give specifically invariant **integers**. This is not obvious from what I have said, but can be proven via understanding De Rahm cohomology groups. With that highly non trivial but useful fact out of the way, first remember that $c_j(F)$ is a $2j$ form. This means it must be integrated over a $2j$ surface. As an example, consider a $U(1)$ principal bundle over a sphere $M = S^2$ and then consider a $\mathbb{C}$ vector bundle associated to it. In this case $Tr(F) = F$ since we are working with $U(1)$. Lets integrate it over M (i.e. a ball around the origin in $\mathbb{R}^3$

$$\begin{aligned} \int_{S^2} c_1 &= \int_{S^2} \frac{i}{2\pi} F = n \\ \Phi_B &= 2\pi n \end{aligned}$$

This is Dirac's quantization condition! One more example, notice that $c_2(F) \sim Tr(F \wedge F)$ for $SU(N)$ since $Tr(F) = 0$. This is precisely the term which when integrated over gives an integer, corresponding to instanton number. This is where

$$k = \frac{1}{8\pi^2} \int Tr(F \wedge F)$$

came from, and now we actually have a rigorous method for calculating the integer which is parameterizing our bundles as I mentioned earlier! Now Im sure that in higher dimensional spacetime there are some other interpretations to what the higher classes mean but idk them.

Now lets take a look back and notice that this polynomial I defined above kind of popped out of nowhere, and that was because the Chern Weyl theorem didn't tell us what P to pick, it just said that if you have a

valid P , then those properties are true and that you can sort them via their cohomology classes. Lets define a different but very similar polynomial called the total Chern character.

$$ch(F) = Tr \exp \left(\frac{iF}{2\pi} \right)$$

We can do the same process and expand it, and find that it is related to the Chern classes via

$$\begin{aligned} ch_0(F) &= k \\ ch_1(F) &= c_1(F) \\ ch_2(F) &= \frac{1}{2}c_1(F)^2 - c_2(F) \\ &\vdots \end{aligned}$$

where k is the largest number such that $k < 2n$ with n being the dimension of M . Equivalently, the trace of I . By construction this should contain the same topological data (technically some torsion stuff is lost here but whatever), just repackaged in a slightly different way. Notice for the above $SU(2)$ example that since $c_1(F) = 0$, then the Chern characters and classes are very closely related. We will see next section where the Chern characters naturally pop up.

We could keep defining other classes, for example the Todd class is something we can define and is useful for writing down the Atiyah-Singer index theorem in general. There is also the Steiffel Whitney class which doesn't use the curvature but helps determine if a manifold admits spin, but I won't delve into those here as I don't have enough time :(.

To recap, characteristic classes are a way of assigning a vector bundle over a space M to an element of the cohomology group $H^*(M, A)$ with A being some abelian group. An element of a cohomology group (called a cohomology class), is associated a value in A . To start computing these we constructed a definition of an invariant polynomial which we found was closed, meaning that it is topological (i.e. an element in the cohomology group), and that it also differs by an exact form when perturbed, so that perturbations stay within the same cohomology class. We found that the Chern classes allowed us to assign our vector bundles an element $H^*(M, \mathbb{Z})$ i.e. an integer after we integrate. Chern characters assign bundles and element $H^*(M, \mathbb{Q})$, i.e. a rational number. Both appear naturally in different contexts. As a side note Ive avoided a strange subtlety. Notice that Chern classes are naturally defined as elements in $H^*(M, \mathbb{Z})$. However, our construction clearly shows that they are elements of the De Rahm cohomology which is isomorphic to $H^*(M, \mathbb{R})$ since differential forms are real (or complex) valued. So what is going on here? Its just a bit of a definitional abuse, there is a homeomorphism that maps $H^*(M, \mathbb{Z}) \rightarrow H^*(M, \mathbb{R})$, so technically this construction is creating differential forms in the image of this homeomorphism so they still integrate to integers. Not a big deal just thought I'd throw that out in case of confusion. Same deal with Chern characters and $\mathbb{Q}$.

5 Differential Operators and the Index Theorem

Now we know how to construct topological invariants, but how can we compute the affect this has in terms of anomalies? The next step towards this is the Atiyah-Singer index theorem.

I think that delving into the details of defining what a differential operator is in terms of bundles will just add too much time that I don't have, but basically its a map between sections over different vector bundles. But lets sort of ignore that formal definition and just think of a differential operator D in the way we normally would in physics. For example, the Dirac operator $\not{D}$ is a differential operator. We can define the analytical index of a differential operator as follows

$$\text{ind}D = \dim \ker D - \dim \ker D^\dagger$$

where remember that $\ker D$ is just the set of fields which obey $D\psi = 0$. We need to restrict our differential operators to a class of them called elliptic. Dont worry about what that means just know they more or less describe static equilibrium. However, its possible for the dimensions of the kernels to be infinite in which computing the analytical index is sort of nonsense. It can be showed that Elliptic operators on a compact space are Fredholm, meaning that these quantities are finite. Remember how we compactified $\mathbb{R}^4$ earlier? Well thats going to be of tremendous help here in that all of our equations are Fredholm, meaning that analytic index is finite and meaningful. The point of the AS index theorem is that calculating the analytic index is incredibly hard, and that its possible to instead compute a topological index, which is equal by the AS index theorem. However the general theorem is sophisticated and is not worth the time going into. Rather we will constrain ourselves to the Dirac operator $\not{D}$ has an analytical index of

$$\text{ind}\not{D} = n_+ - n_-$$

where $n_\pm$ is the number of right/left handed zero modes. Please note that I am skipping over a lot of formality just to get to the punchline. But understand that to make sense of spin you have to take your associated vector bundle and add to it a Spin bundle which splits into a left and right handed piece. This setup makes the AS index theorem much nicer, and for the Dirac operator we get

$$n_+ - n_- = \int_M \hat{A}(R) \wedge ch(F)$$

where $\hat{A}(R)$ is called the Dirac genus (R is the curvature of spacetime). Fun fact, this physics version can be proved using SUSY! All that really matters is that in flat spacetime, where $R = 0$, we have $\hat{A} = 1$ (however if we wanted to we could do Dirac stuff on any spacetime, it would just get complicated cause of $\hat{A}$). So that means

$$n_+ - n_- = \int_M ch(F)|_{vol}$$

where by *vol* I am just clarifying that we only care about the piece of the expansion of the Chern character that is the same dimension of our spacetime. Wow! This is incredible, we can compute something purely topological and then relate that to the analytic index of our differential operator! So in 2D (Schwinger model) we would integrate $ch_1(F)$, and for normal spacetime we would integrate $ch_2(F)$. Also, while I haven't quite shown the connection between this and anomalies quite yet, notice that nothing interesting happens in odd dimension since there is no Chern character of odd order, as a result we can maybe see why people say "there is no chiral anomaly in odd dimension"!

6 The Chiral Anomaly

Lets quickly remind ourselves of what Cameron showed us earlier in the quarter. Fujikawa showed us that when we take our fermionic path integral

$$Z = \int \mathcal{D}\psi \mathcal{D}\bar{\psi} e^{-\int dx \bar{\psi} \not{D} \psi}$$

and perform a local chiral transformation by $e^{i\gamma_5 \alpha(x)}$, that the path integral measure contains a Jacobian factor which is nontrivial and is given by

$$J^{-2} = e^{-2i \int dx \alpha(x) \sum \psi_n^\dagger \gamma_5 \psi_n}$$

where the ψ_n 's are just eigenvectors of $\not{D}$ with eigenvalue λ_n . And so we defined the anomaly function

$$A(x) = \sum \psi_n^\dagger \gamma_5 \psi_n$$

and since α was arbitrary we were left with

$$\partial_\mu j_A^\mu = -2iA(x)$$

Cool. But what does topology have to do with this? Take a step back, lets consider α to be constant, and then notice that in the Jacobian factor the integral

$$\int dx A(x) = \int dx \sum \psi_n^\dagger \gamma_5 \psi_n$$

is UV divergent. So, lets include a heat kernel regulator (which is manifestly gauge invariant)

$$\begin{aligned} \int dx \sum \psi_n^\dagger \gamma_5 \psi_n &\rightarrow \int dx \sum \psi_n^\dagger \gamma_5 \psi_n e^{-(\lambda_n/M)^2} \\ &= \sum \langle \psi_n | \gamma_5 e^{-(i\not{D}/M)^2} | \psi_n \rangle \end{aligned}$$

where we will send $M \rightarrow \infty$. What you can show by just simply using the Dirac equation is that the only contribution to this sum is the zero modes (so that M goes away even before we take any limit), i.e.

$$\int dx A(x) = n_+ - n_-$$

but wait a second, the AS index theorem tell us that $n_+ - n_- = \int ch(F)|_{vol}$. So that just means that

$$A = ch|_{vol}$$

In 4D, then

$$A = ch_2(F) = -\frac{1}{8\pi^2} Tr(F \wedge F)$$

$$\partial_\mu j_A^\mu = \frac{i}{4\pi^2} Tr(F \wedge F) = \frac{i}{8\pi^2} Tr(F_{\mu\nu} \tilde{F}^{\mu\nu})$$

This is exactly the chiral anomaly! Ok, but was it worth 10 pages of topology and differential geometry to get the same result we got using much easier methods. Well yes, there are two major upshots to this. Triangle diagrams and perturbative methods compute the anomaly at one loop. It has been shown that higher loop corrections are 0, but that is rather unclear. Instead, this is global and completely non perturbative from the start, we can be confident that it is the exact answer. Secondly, this was one application of this procedure, I could have stepped back and made my spacetime curved, or made it some other funky compact space and tackling that with this formalism is 1000 times easier. You can also derive the non abelian anomalies I just don't really have time.

7 Descent Equations, Non Abelian Anomalies, and Gauge Anomalies

Would have liked to talk about this but didn't really have time to :/. Id read chp. 13 of Nakahara if interested.

8 Conclusion

We now have a topologically rigorous way of computing anomalies, perturbative or not, using results from topology and the index theorem. Note that if we had a way to compute the entire path integral we wouldn't ever have to worry about this, but it is because we are forced to use perturbation theory to do analytic computations, that we then have to reconcile the consequences that come with doing that. Lattice QCD allows us to literally calculate the entire path integral, and its why lattice QCD calculations see the anomalies.